

TODO title about Y-haplotype effect on larval transcriptome

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Sample numbers

- Day 14
 - Small-Y: 10 males, 3 females
 - Large-Y: 9 males, 6 females
- Day 16
 - Small-Y: 11 males, 3 females
 - Large-Y: 7 males, 6 females
- Day 18
 - Small-Y: 7 males, 5 females
 - Large-Y: 12 males, 3 females

Additional results

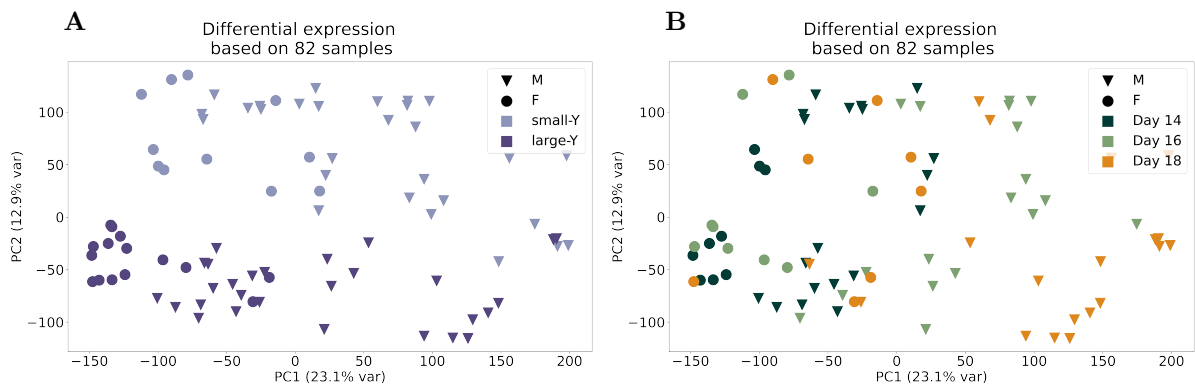


Figure S1: **PCA plots of full dataset.** (A, B) PCA of all samples, highlighted by Y-haplotype (A) or time point (B).

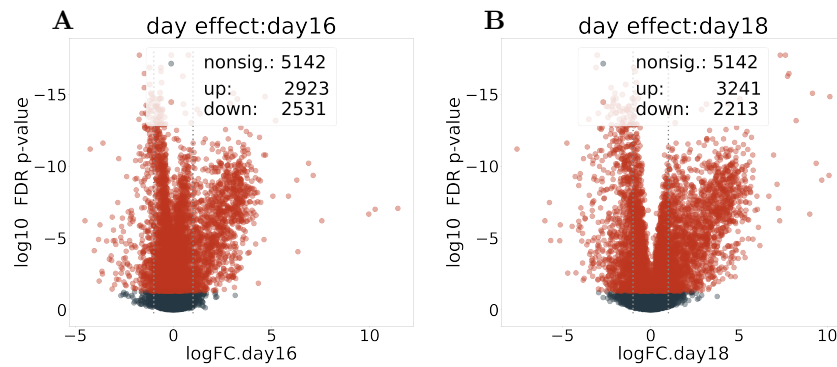


Figure S2: **Volcano plots of day-comparisons on full dataset.** The comparison is of coefficients day 16 (A) and day 18 (B) to the intercept (day 14).

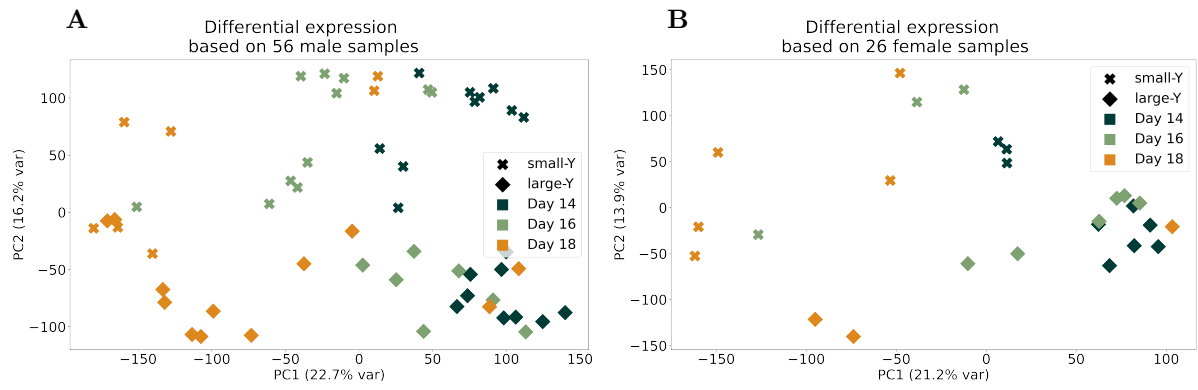


Figure S3: **PCA plots of male and female samples separated.** Separate analysis of male (A) and female (B) samples.

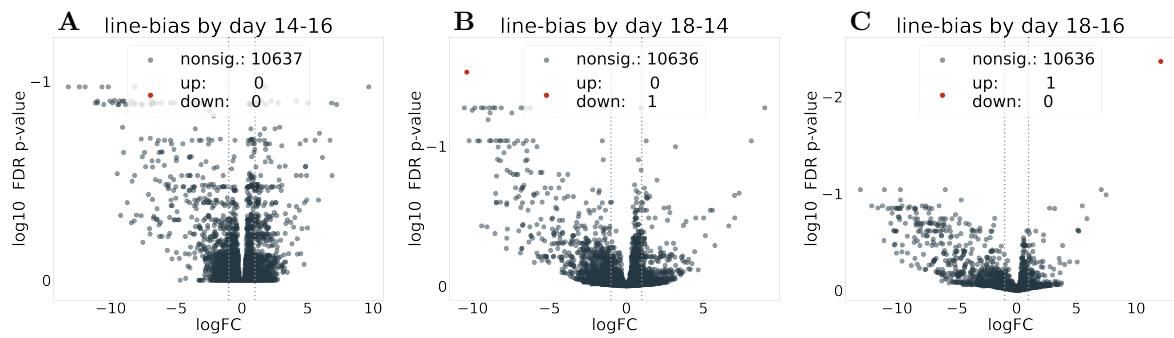


Figure S4: **Volcano plot of male samples line-by-day interaction.** The interaction is calculated between the contrasts of smallY-largeY and each pairwise day comparison, day 14-16 (A), day 18-14 (B) and day 18-16 (C).

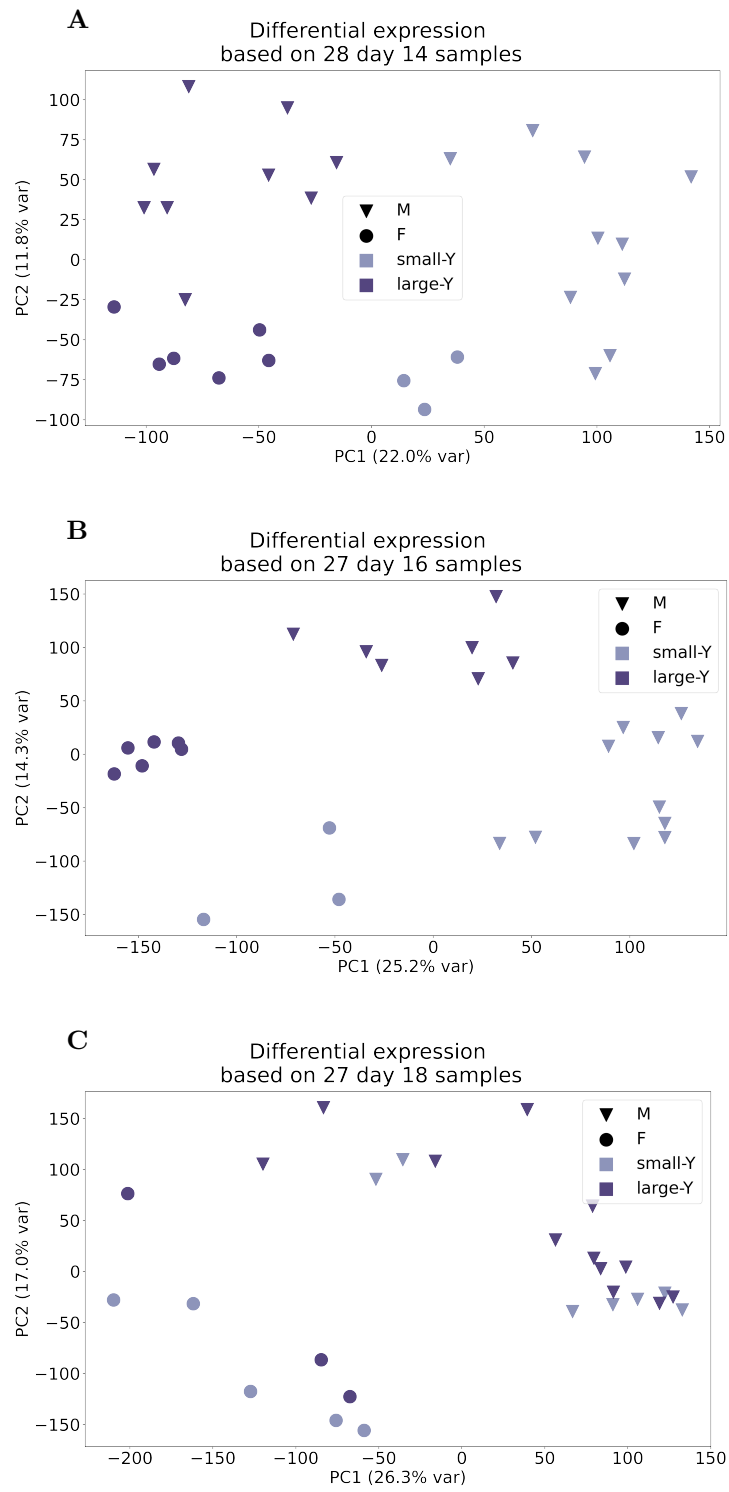


Figure S5: **PCA plots of samples from each time point separately.**

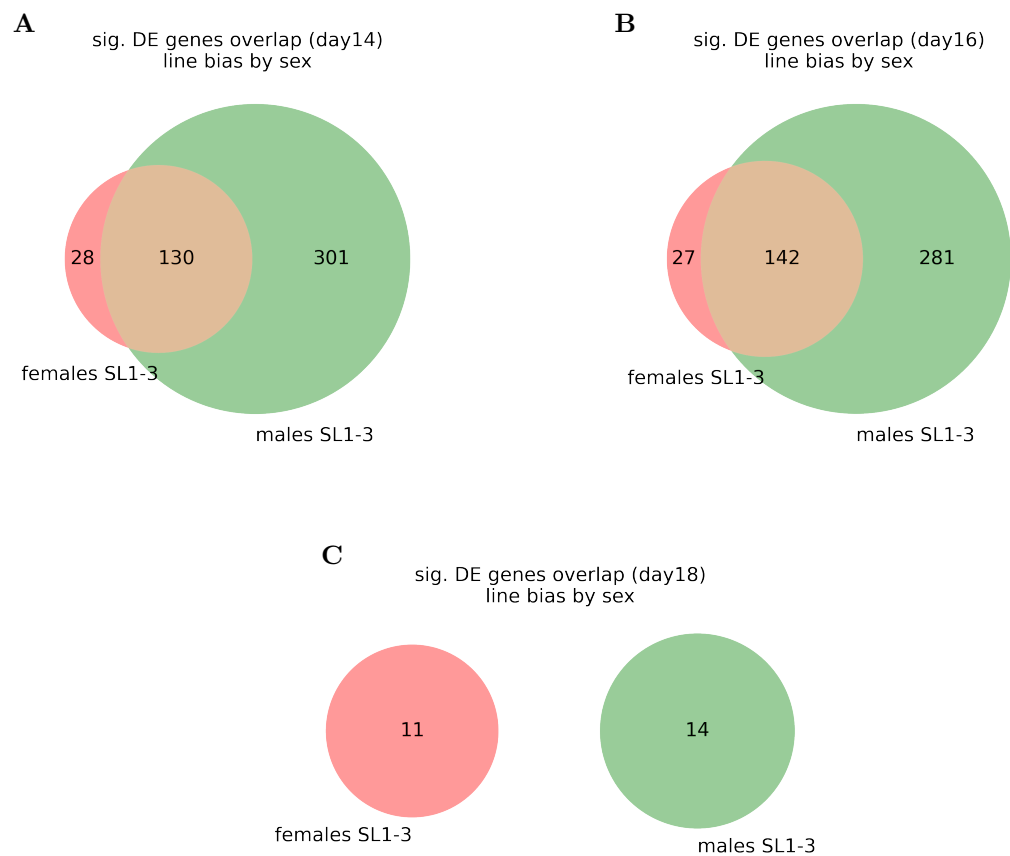


Figure S6: **Overlap of line-biased genes between the sexes.** One overlap for each day, significance calculated on day-separated samples.

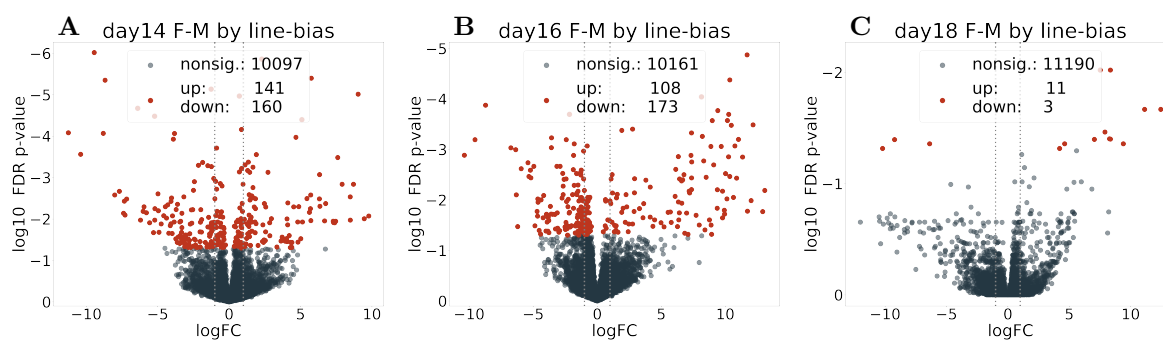


Figure S7: **Volcano plot of line-by-sex interaction for each time point.**

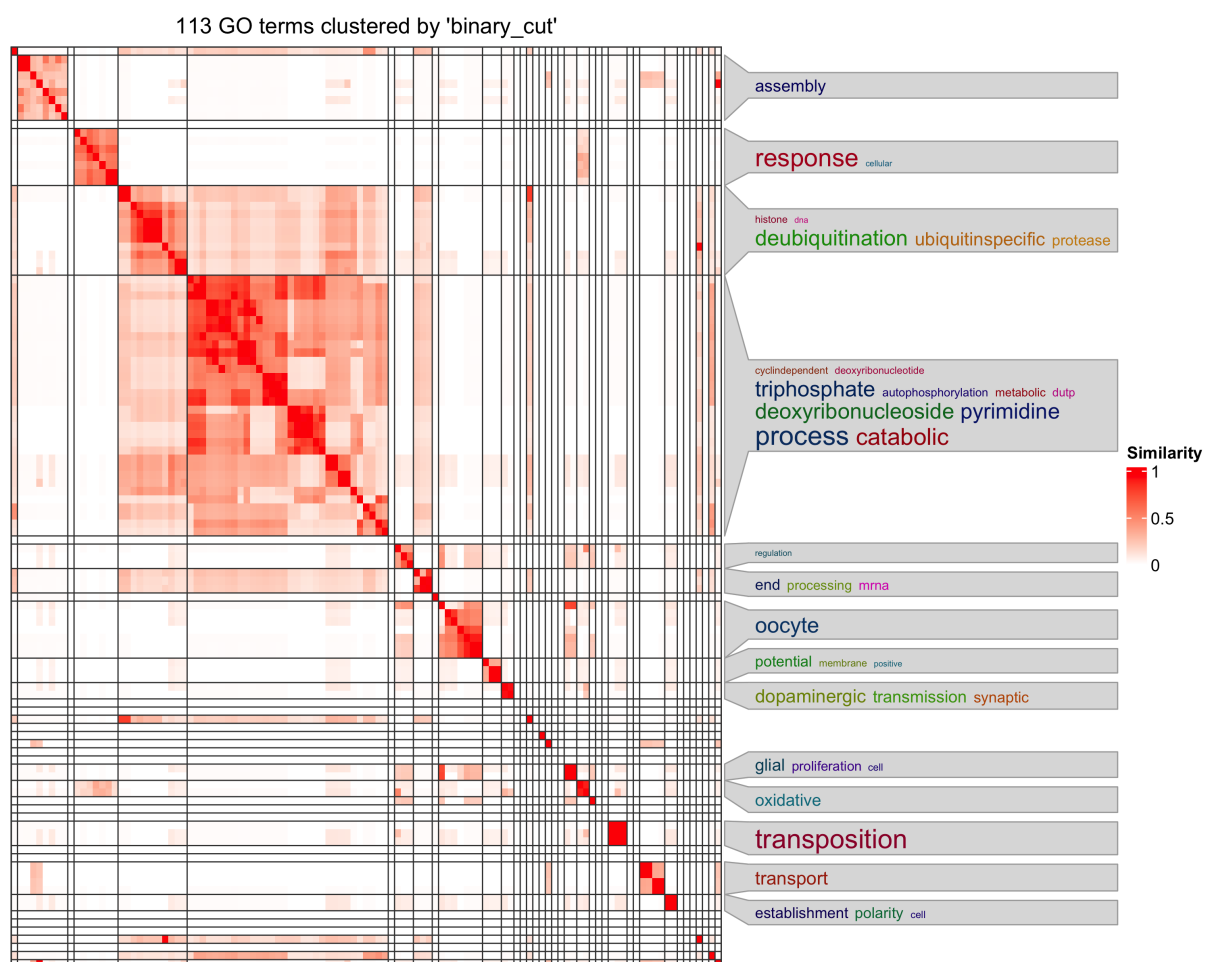


Figure S8: GO-term enrichment and semantic clustering of male line-biased samples on day 14.

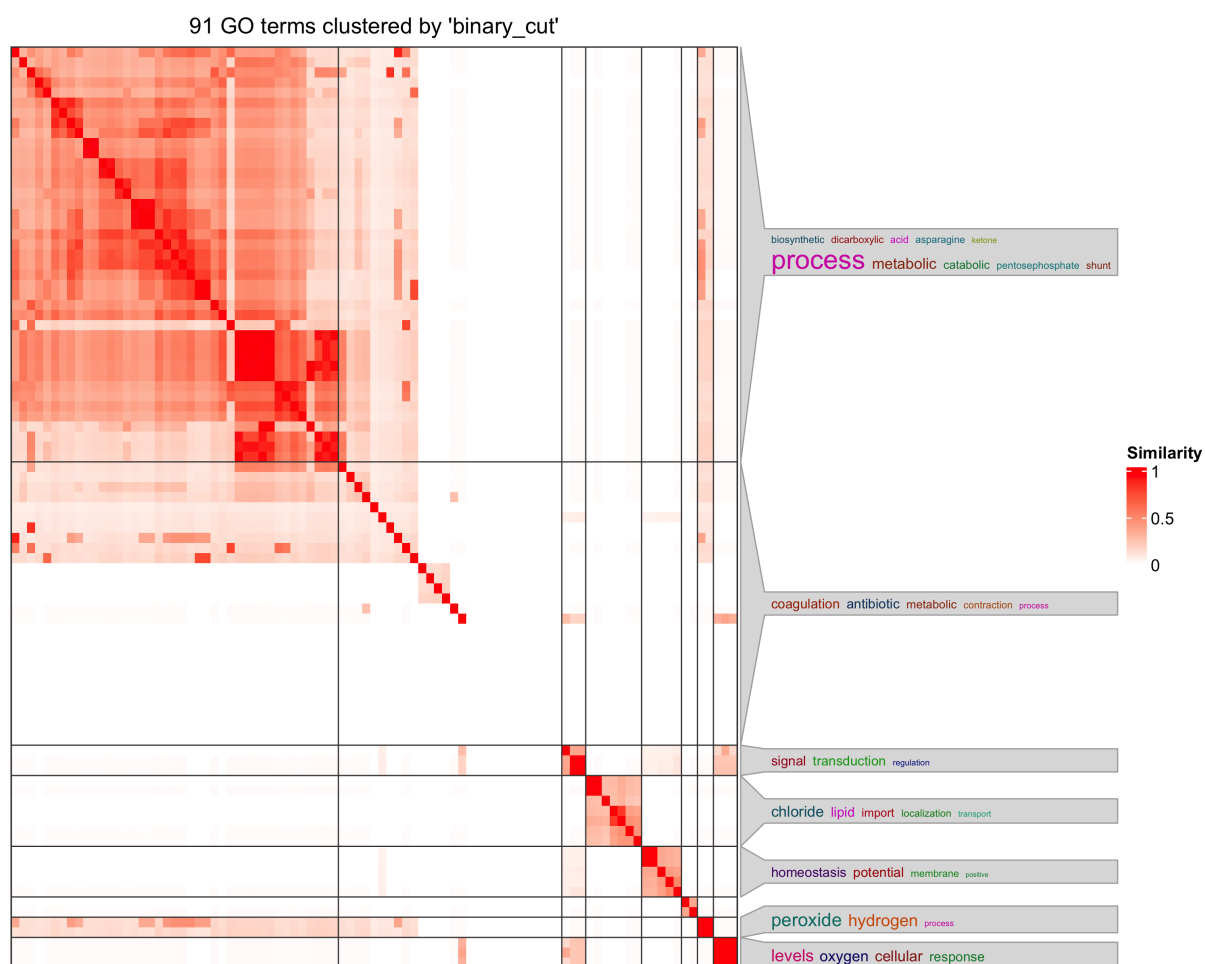


Figure S9: GO-term enrichment and semantic clustering of male line-biased samples on day 16.



Figure S10: **GO-term enrichment and semantic clustering of male line-biased samples on day 18.**

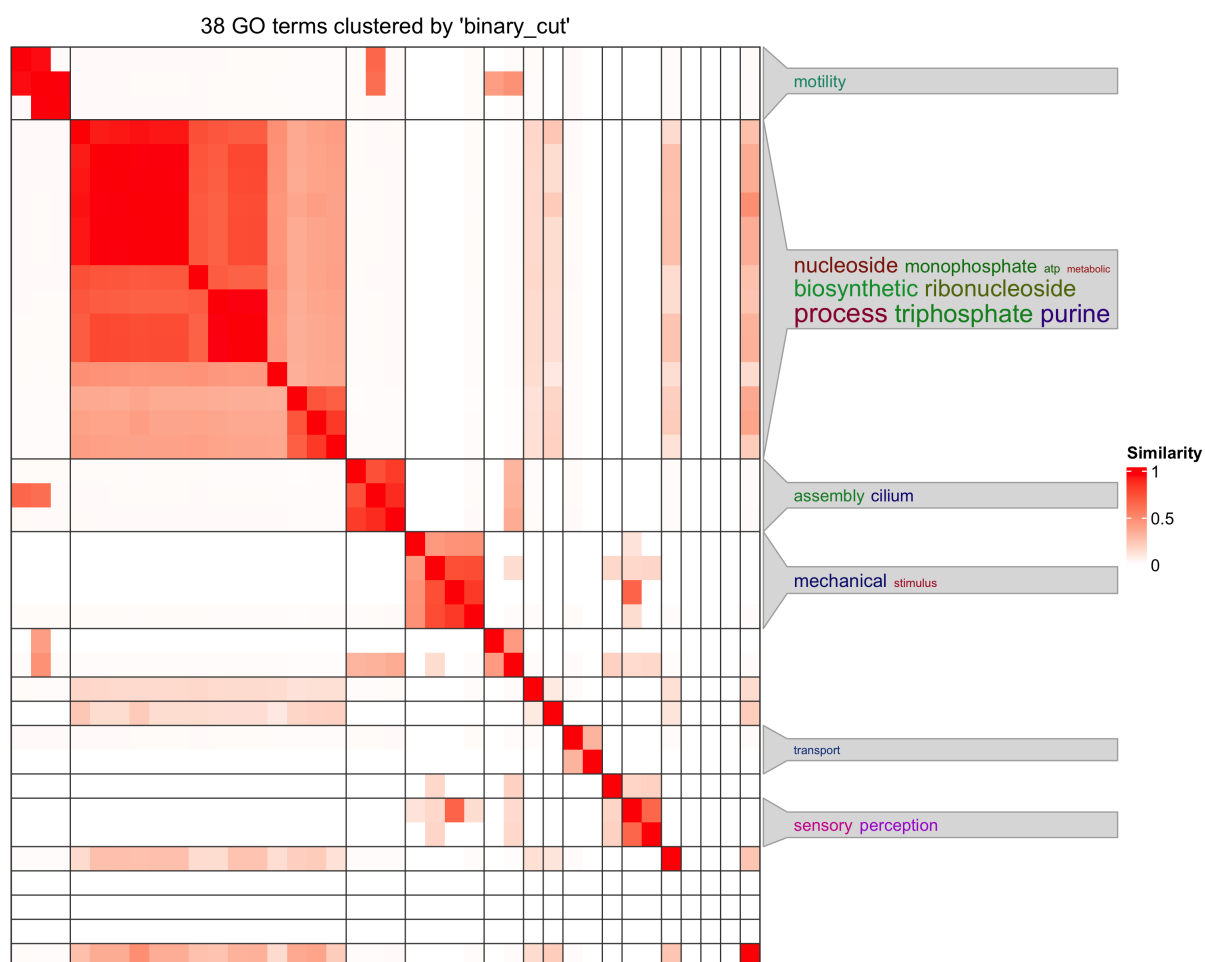


Figure S11: **GO-term enrichment and semantic clustering of genes that are sex-biased.** Full dataset with line as fixed effect, GO terms that are enriched in sex-biased genes of all three time points, intersection as specified in Fig. 2D

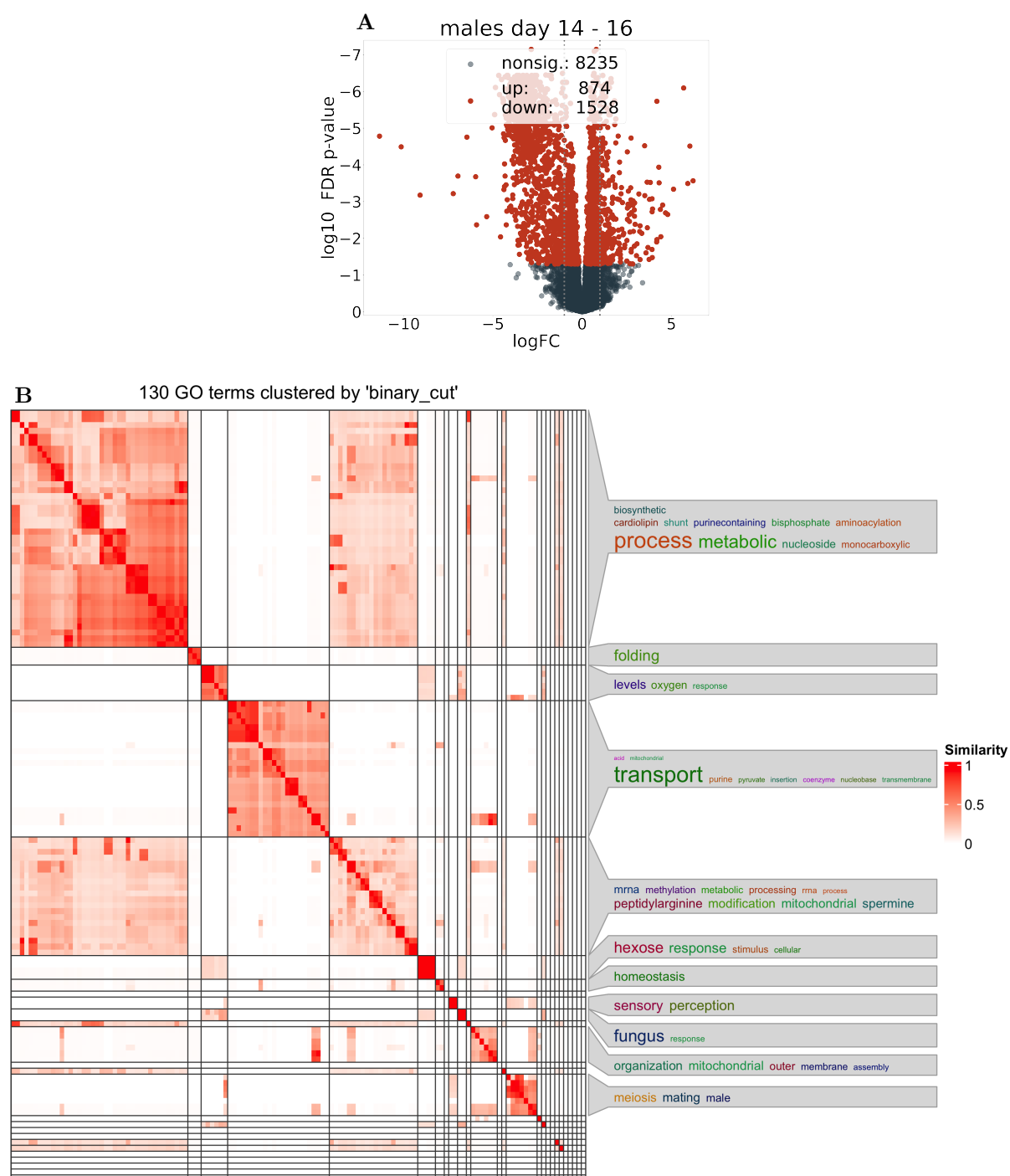


Figure S12: **volcano plot, GO-term enrichment and semantic clustering of genes that are sex-biased.** Only male samples with day 14-16 comparison (Fig. 2E intersection). (A) shows the volcano plot, and (B) the semantic clustering of GO-terms that are enriched in sig. DE genes.

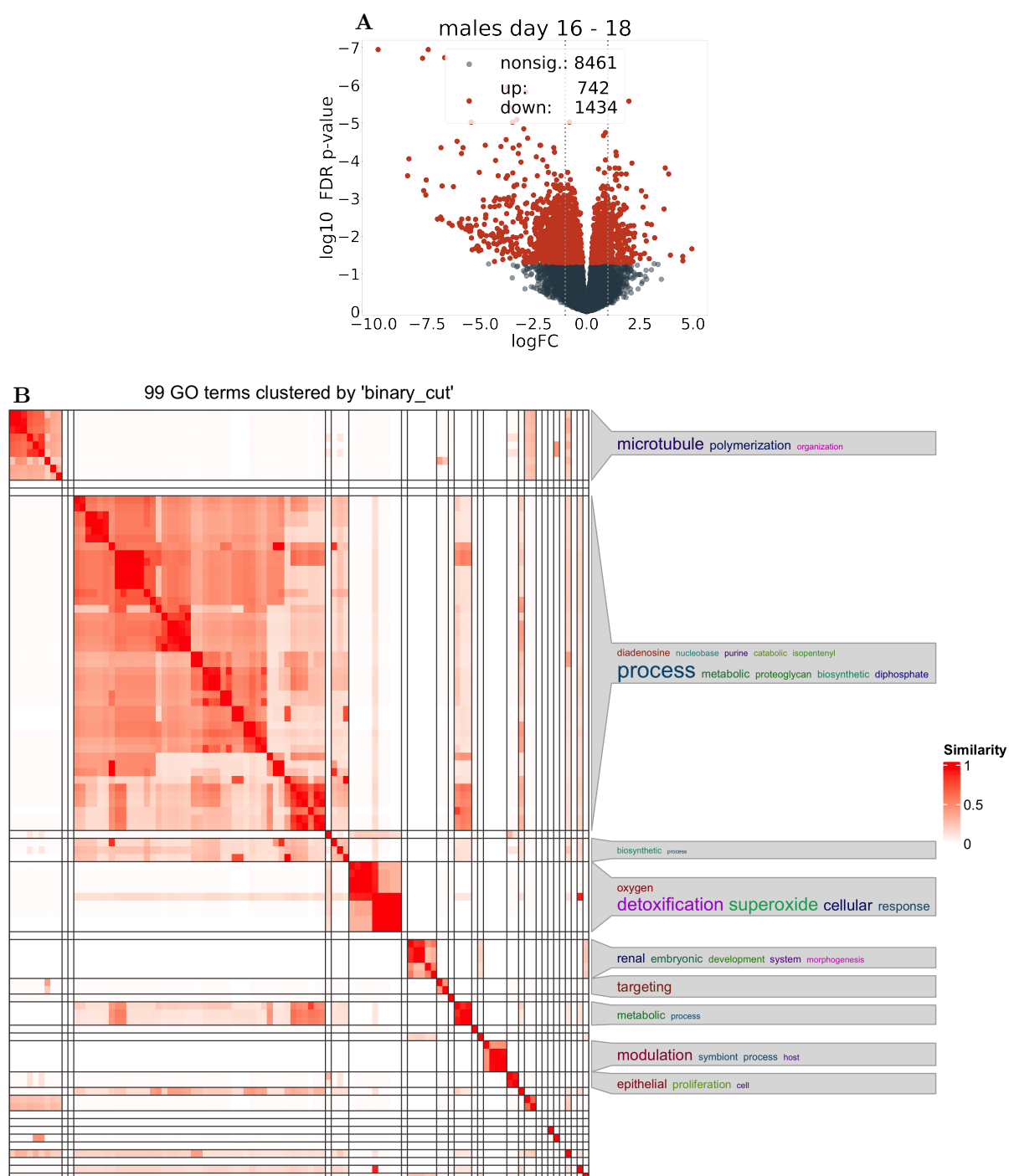


Figure S13: **volcano plot, GO-term enrichment and semantic clustering of genes that are sex-biased.** Only male samples with day 16-18 comparison (Fig. 2E intersection) (A) shows the volcano plot, and (B) the semantic clustering of GO-terms that are enriched in sig. DE genes.

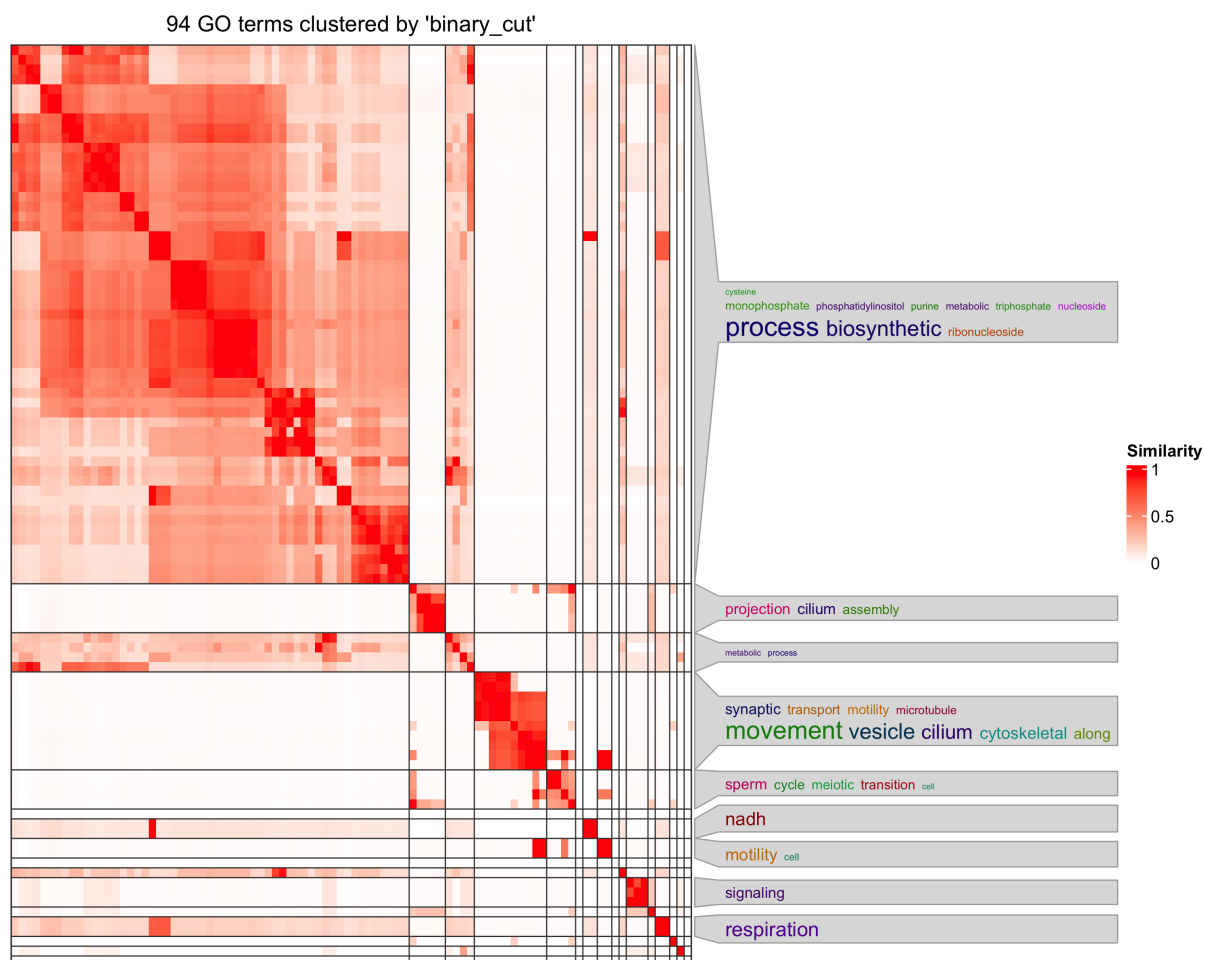


Figure S14: **GO-term enrichment and semantic clustering of genes that are sex-biased on all days.** Sex-bias is estimated separately for each time point in the day-split data, genes that are sex-biased in both lines and all days are determined and GO-enrichment and semantic clustering is performed.

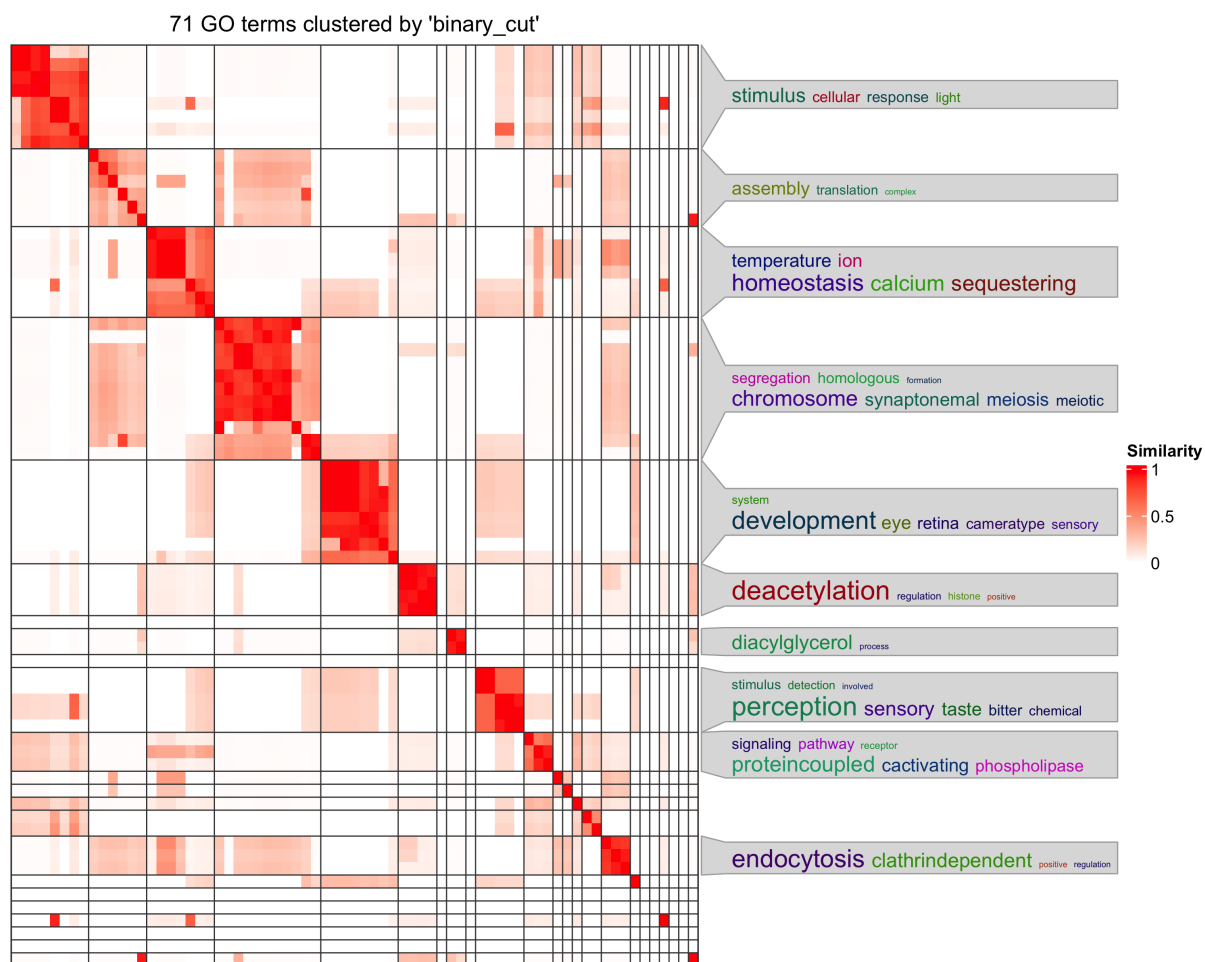


Figure S15: **GO-term enrichment and semantic clustering of genes that are sex-biased on day 14.** Sex-bias is estimated separately for each time point in the day-split data, genes that are sex-biased only on day 14 are determined and GO-enrichment and semantic clustering is performed.

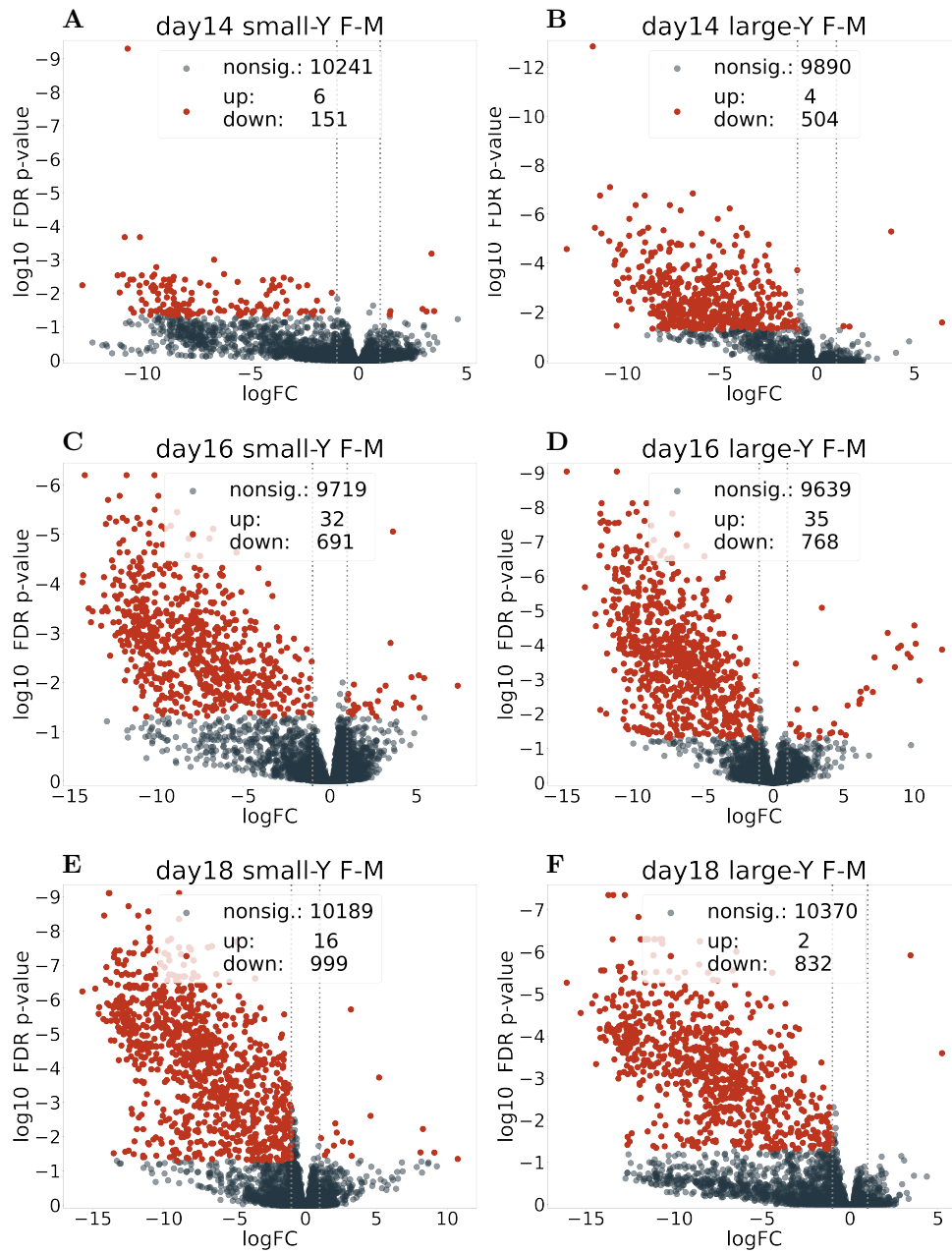


Figure S16: Volcano plot of sex-bias within each line for samples of each time point.

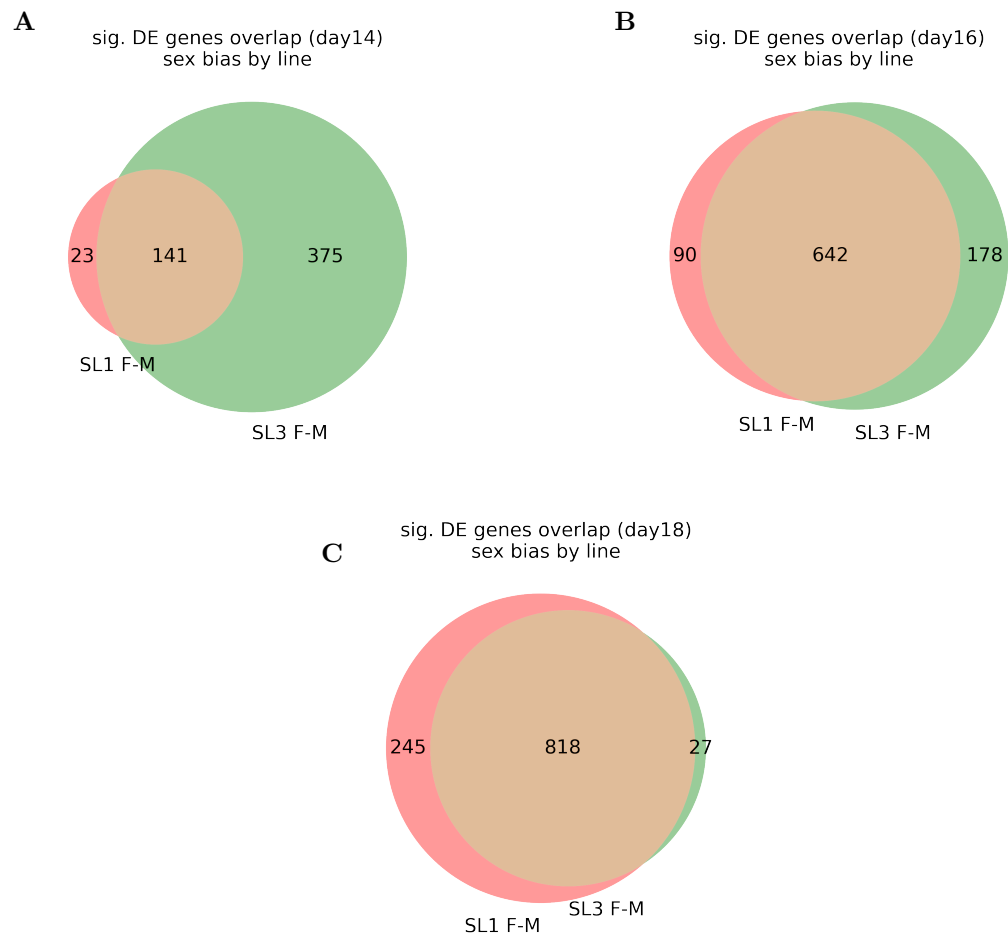


Figure S17: **Overlap of sex-biased genes between the lines.** SL1 refers to small-Y males and SL3 refers to large-Y males.

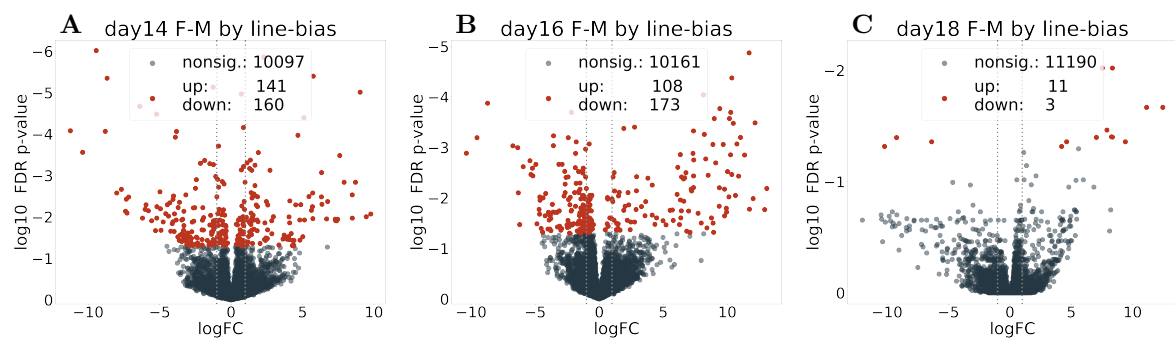


Figure S18: **Volcano plot of line-by-sex interaction for samples of each time point.**

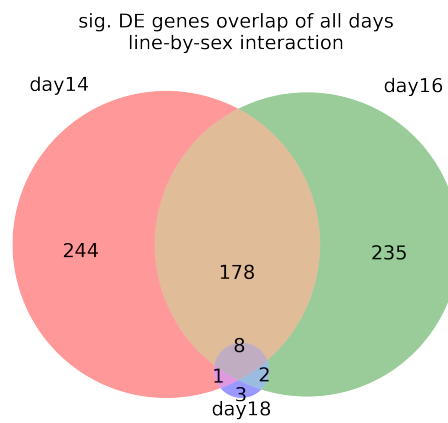


Figure S19: **Overlap of genes significant in the line-by-sex interaction between all days.**

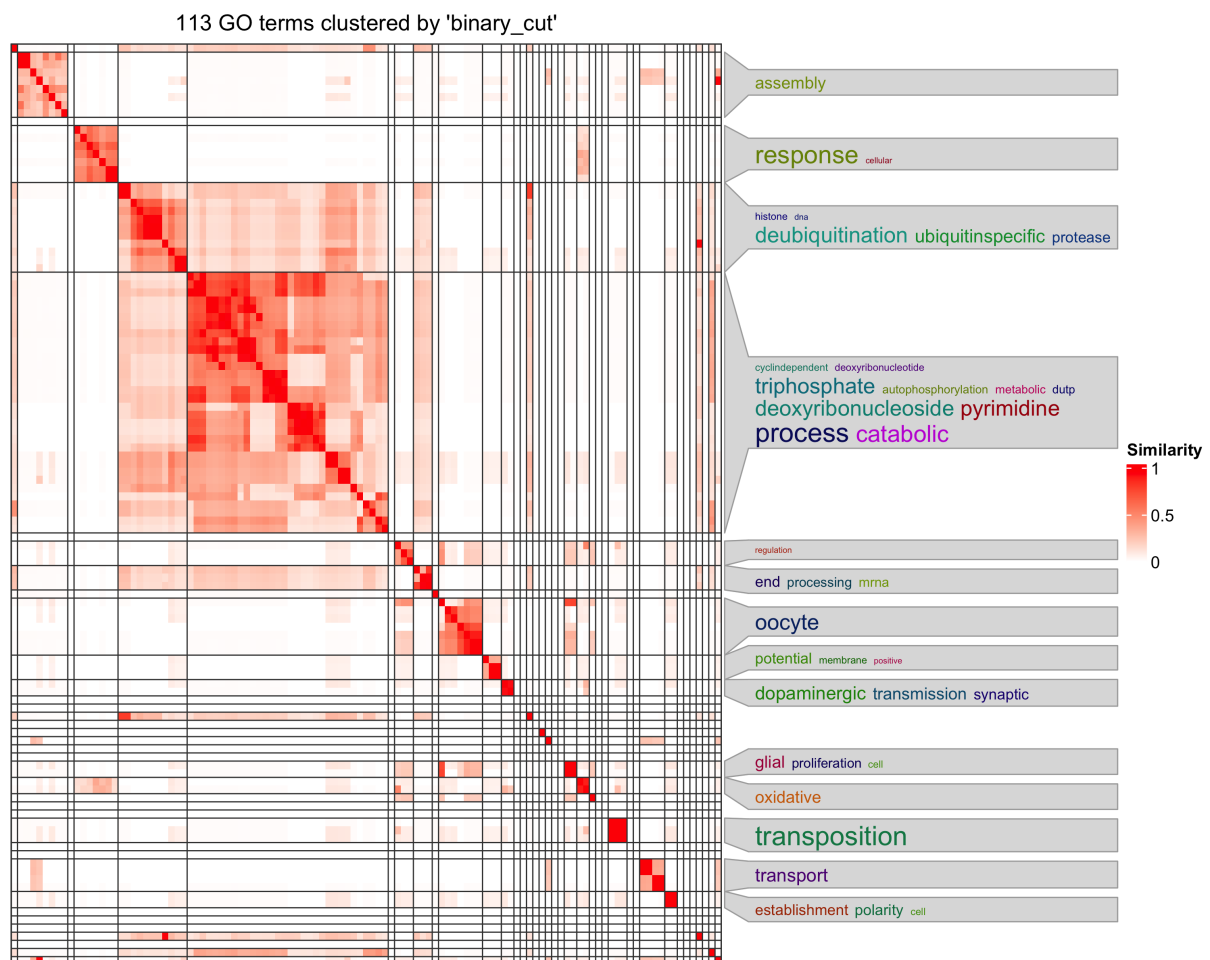


Figure S20: **GO-term enrichment and semantic clustering of genes that significant in the line-by-sex interaction on day 14**

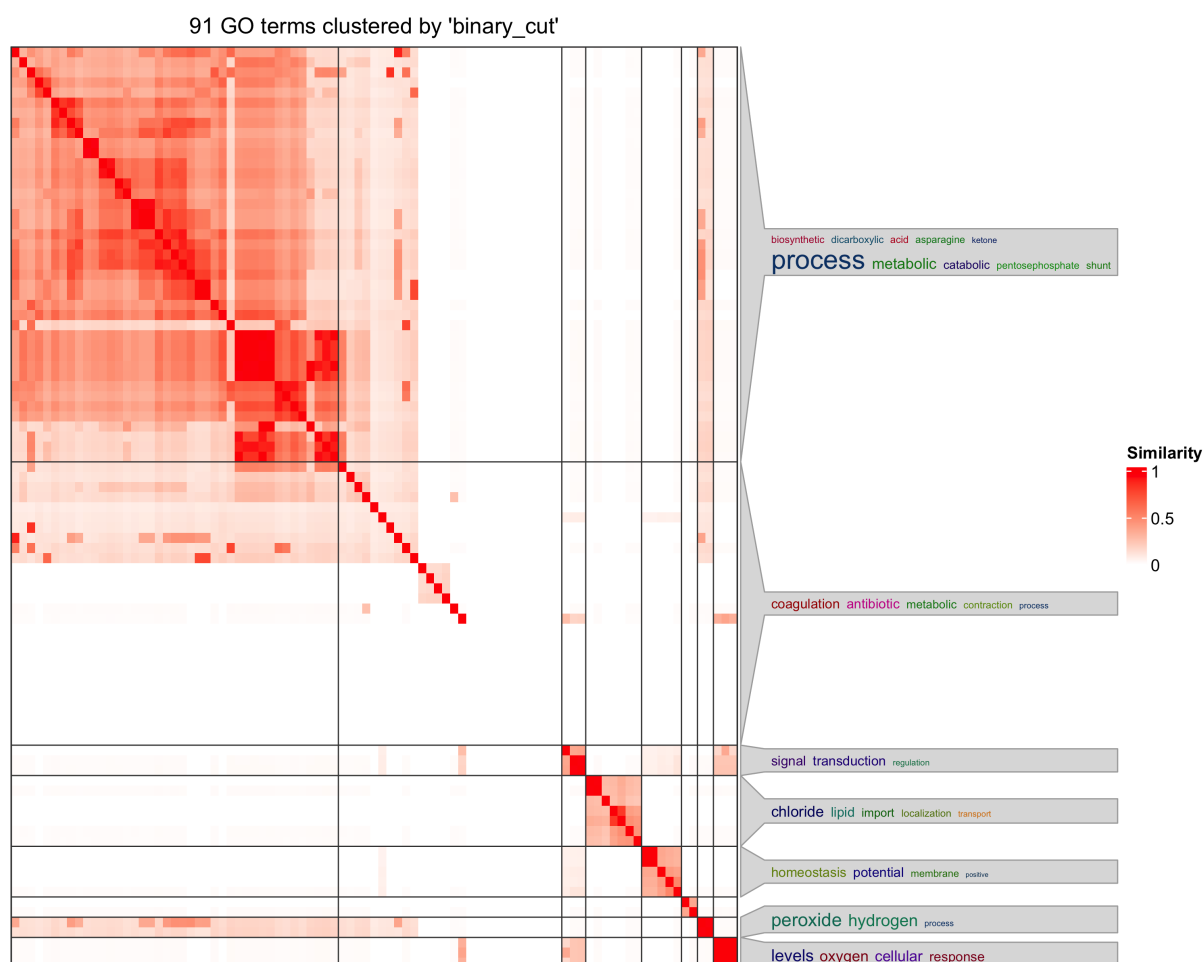


Figure S21: **GO-term enrichment and semantic clustering of genes that significant in the line-by-sex interaction on day 16**

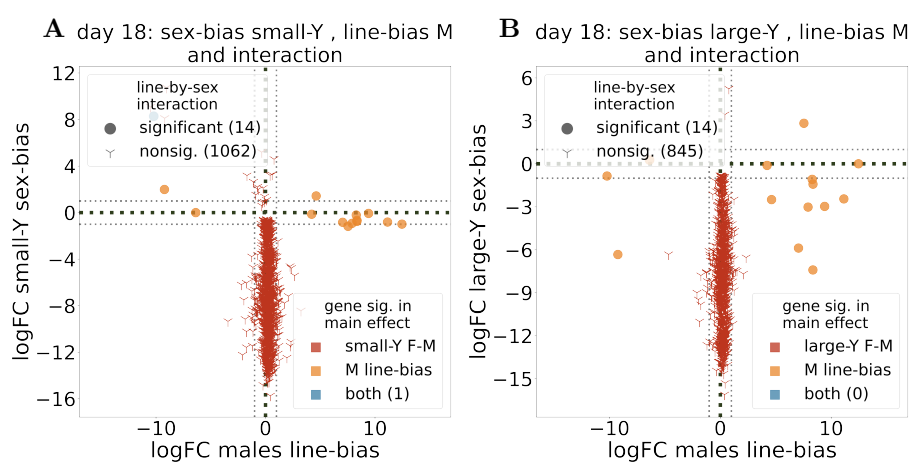


Figure S22: **Intersection of sex bias and line bias across developmental stages day 18. Log₂FC of sex-bias (females - males contrast) and line-bias (large-Y - small-Y) of day 18.**



Figure S23: **GO-term enrichment and semantic clustering of genes that significant for sex-bias in large-Y lines and line-bias in males on day 14.** The background of this is genes that are significant in large-Y sex bias, male line-bias or the sex-by-line interaction on day 14 (all genes shown in Fig. 4B).

Gene ID	Description
gene-149137	no_description
gene-372264	—not_annotated—
gene-227137	no_description
gene-224956	Ribosomal L37ae protein family
gene-414353	no_description
gene-39545	Protein of unknown function (DUF1676)
gene-115312	Transferrins are iron binding transport proteins which bind Fe(3) ion in association with the binding of an anion, usually bicarbonate
gene-130081	no_description
gene-371957	serine-type endopeptidase activity. It is involved in the biological process described with proteolysis
gene-204669	Lipase
gene-166391	DM4/DM12 family
gene-246615	—not_annotated—
gene-277078	G-protein coupled receptor activity. It is involved in the biological process described with G-protein coupled receptor signaling pathway
gene-406603	Ciliary basal body-associated, B9 protein
gene-158197	—not_annotated—
gene-122220	Belongs to the short-chain dehydrogenases reductases (SDR) family
gene-100036	Zinc-dependent metalloprotease
gene-143368	Zinc carboxypeptidase
gene-130096	no_description
gene-398993	Src homology 3 domains
gene-220249	axonemal central apparatus assembly

Table S1: Gene IDs and annotated functions (from eggnog) of genes sig. in both (blue) in Fig. 4B.

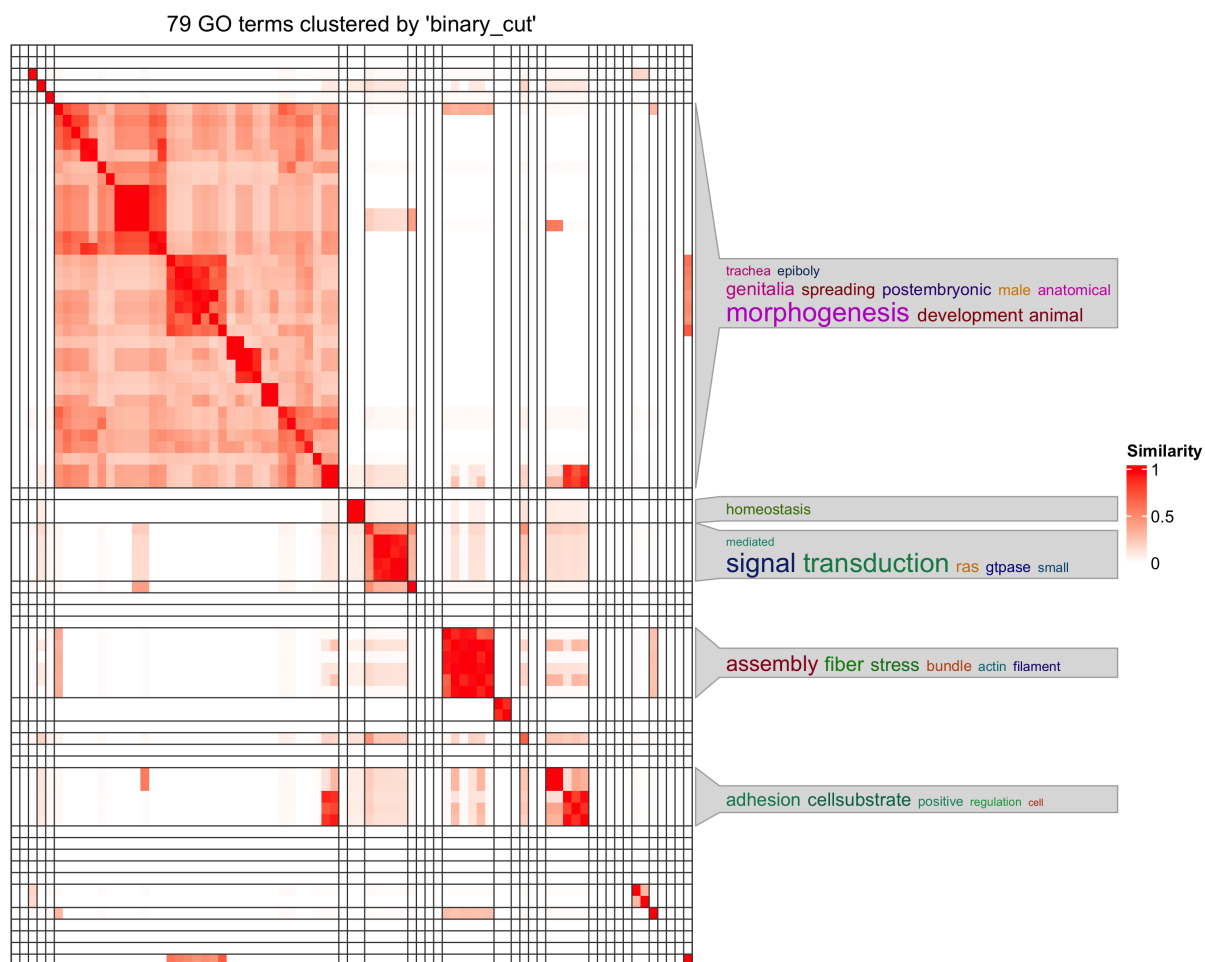


Figure S24: **GO-term enrichment and semantic clustering of genes that significant for sex-bias in large-Y lines and line-bias in males on day 16.** The background of this is genes that are significant in large-Y sex bias, male line-bias or the sex-by-line interaction on day 16 (all genes shown in Fig. 4D).

Gene ID	Description
gene-372264	—not_annotated—
gene-23413	Belongs to the ubiquitin-activating E1 family
gene-24120	RNA polymerase II transcription cofactor activity. It is involved in the biological process described with regulation of transcription from RNA polymerase II promoter
gene-224357	Peptidase S24-like
gene-15763	7 transmembrane receptor (rhodopsin family)
gene-23365	Belongs to the TRAFAC class myosin-kinesin ATPase superfamily. Myosin family
gene-349163	Reverse transcriptase (RNA-dependent DNA polymerase)
gene-24203	Leucine-rich repeat-containing protein
gene-24221	C-terminal domain on Strawberry notch homologue
gene-24167	Transport and Golgi organisation 2
gene-23689	Formin Homology 2 Domain
gene-225030	Belongs to the heat shock protein 70 family
gene-12075	—not_annotated—
gene-24290	Conserved hypothetical ATP binding protein
gene-410366	no_description
gene-24052	Pfam:DUF3588
gene-423321	—not_annotated—
gene-23514	Belongs to the ubiquitin-activating E1 family
gene-24079	—not_annotated—
gene-24088	Mob1/phocein family
gene-7220	—not_annotated—
gene-24132	Guanylylate cyclase
gene-23840	FKBP-type peptidyl-prolyl cis-trans isomerase
gene-23597	Myosin tail
gene-24185	Phosphotransferase activity, for other substituted phosphate groups. It is involved in the biological process described with phospholipid biosynthetic process
gene-23538	twin BRCT domain
gene-27466	GTP binding. It is involved in the biological process described with
gene-220249	axonemal central apparatus assembly

Table S2: Gene IDs and annotated functions (from egglog) of genes sig. in both (blue) in Fig. 4D.